

Member ID: _____

Time: _____

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COMPUTER NETWORK TECHNOLOGY (300)

NATIONAL 2022

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each)

_____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Multiple Choice: Identify the letter of the choice that *best* completes the statement or answers the question.

1. Which of the following abbreviations are commonly used to represent transmit and receive?
 - a. Tr and Re
 - b. Tn and Rn
 - c. Tx and Rx
 - d. Ti and Rn
2. What action does the technology, TCP offload engine (TOE), perform?
 - a. Offloads processing of the TCP/IP stack to the network controller.
 - b. Offloads processing of the TCP/IP stack to the processor.
 - c. Offloads TCP packets to be processed by a software controller.
 - d. Offloads UDP packets to be processed by a software controller.
3. Which of the following payload sizes would *not* be considered a jumbo frame / packet?
 - a. 8960 bytes
 - b. 7895 bytes
 - c. 1460 bytes
 - d. 5024 bytes
4. Reference the following shared folder name / path and identify the domain, server, and shared folder names: \\FS01.BPA.org\Volunteering
 - a. Domain: FS01
Server: BPA.org
Share: Volunteering
 - b. Domain: BPA.org
Server: FS01
Share: Volunteering
 - c. Domain: BPA.org
Server: Volunteering
Share: FS01
 - d. Domain: Volunteering
Server: FS01
Share: BPA.org
5. Which of the following names is most commonly used as the DNS suffix?
 - a. Server name
 - b. Domain name
 - c. NetBIOS name
 - d. Network adapter name

6. The file-sharing protocol, NFS, is primarily used by which of the following clients?
 - a. Microsoft
 - b. MacOS
 - c. Android
 - d. Linux
7. The file-sharing protocol, AFP, is primarily used by which of the following clients?
 - a. Microsoft
 - b. MacOS
 - c. Android
 - d. Linux
8. The file-sharing protocol, SMB, is primarily used by which of the following clients?
 - a. Microsoft
 - b. MacOS
 - c. Android
 - d. Linux
9. A network administrator is troubleshooting an issue in which client VPN connections are being disconnected, most commonly when the client machines go into hibernation. Which of the following is the most likely culprit?
 - a. Wait for link.
 - b. Wake on magic packet.
 - c. Allow this device to wake the computer.
 - d. Allow the computer to turn off this device to save power.
10. Which of the following problem statements would the system event log be most likely to help diagnose / resolve?
 - a. A third-party application is crashing when launched.
 - b. An access denied error is given when attempting to open a shared folder.
 - c. A service is failing to start during boot.
 - d. An external website is inaccessible.
11. A user is attempting to access an external website via HTTPS. The browser is reporting that the certificate is invalid; however, the website was not reporting any issues the day before. The user has verified that no changes have been made to the certificate. Which of the following is the most likely culprit?
 - a. The root certificate is not trusted.
 - b. The certificate has expired.
 - c. The hostname does not match what is listed on the certificate.
 - d. The certificate or a certificate in the chain is on the CRL.

12. Which of the following DNS records would allow clients to resolve www.bpa.org and `remote.bpa.org` to the same IP address?
- a. *.BPA.org
 - b. Any.BPA.org
 - c. Wildcard.BPA.org
 - d. \$_.BPA.org
13. Which of the following sets of IP addresses would *not* require a gateway to communicate?
- a. 10.0.0.10/8 and 100.0.0.10/8
 - b. 172.16.0.10/16 and 172.32.0.100/16
 - c. 192.168.10.10/24 and 192.168.100.100/24
 - d. 169.254.0.10/24 and 192.168.254.10/16
14. Which of the following is not considered a bare metal hypervisor?
- a. VMware ESXi
 - b. Xen server
 - c. Hyper-V
 - d. Oracle VM VirtualBox
15. Which of the following is *true* in regards to containers and / or virtual machines?
- a. Containers have an operating system.
 - b. Virtual machines do not have an operating system.
 - c. Containers do not run on top of a hypervisor.
 - d. Virtual machines run on top of a hypervisor.
16. Which of the following is an open platform that is used to package and run applications in isolated environments, known as containers?
- a. Hyper-V
 - b. Docker
 - c. LXC
 - d. VMware
17. What is the purpose of having a quorum, with regards to a failover cluster?
- a. Provides a shared folder for transferring files between nodes.
 - b. Helps prevent split-brain scenarios.
 - c. Facilitates live migration between cluster nodes.
 - d. There is no advantage of having a quorum.
18. Which of the following is known as an open-source container-orchestration system?
- a. Kubernetes
 - b. Microsoft azure
 - c. Ansible
 - d. Amazon web services

19. Which of the following is a valid troubleshooting step for the following error, when logging into an Active Directory domain: “There are currently no logon servers available to service the logon request”?
- Ensure that the client is on Wi-Fi, not Ethernet.
 - Ensure that the client has a static IP address.
 - Ensure that the client has a valid primary DNS server.
 - Ensure that the client has been rebooted at least 3 times.
20. A network administrator is troubleshooting an issue in which client machines intermittently cannot reach external websites and when external websites are reached, it seems very slow. Accessing websites via IP address does not have this issue. Which of the following is the most likely culprit?
- DHCP
 - Active Directory
 - APIPA
 - DNS Forwarder
21. Which portion of the following MAC address is used to identify the manufacturer / vendor:
70-85-C2-AB-86-42
- 70-85-C2
 - 70 and 42
 - AB-86-42
 - 85-C2-AB-86
22. Which of the following will likely fail to connect when accessing the website for BPA.org?
- http://BPA.org
 - 443://BPA.org
 - https://BPA.org
 - https://BPA.org:443
23. Which of the following is *not* a valid type of virtual switch within Hyper-V?
- External network
 - Internal network
 - Private network
 - Distributed network
24. Which of the following is a *likely* culprit for the error message, “Network path cannot be found”, when attempting to access a shared folder?
- The shared folder must be mapped to a drive letter.
 - The hostname specified does not resolve to a valid IP address.
 - The shared folder is empty.
 - An IP address was specified, instead of the hostname.

25. A network administrator changed the default listening port of Remote Desktop Protocol to 3390 on the Remote Desktop server (hostname: RDServer, IP address: 192.168.1.10). When specifying the Remote Desktop server and port number through RDP, which of the following is not considered valid?
- RDServer:3390
 - 192.168.1.10:3390
 - 3390:RDServer
 - 192.168.1.1:2323
26. Which of the following would not be a symptom that is seen when a broadcast storm occurs?
- Network performance degradation
 - Intermittent packet loss
 - Large number of packets from a single device
 - Successful packet delivery
27. What is one of the primary advantages of using a small form-factor pluggable (SFP)?
- Prevents ports from causing a loop
 - Allows bridging between multiple ports
 - Allows ports to be equipped with different types of transceivers
 - Prevents jumbo frames / packets from exceeding ~9000 bytes
28. Which of the following are valid fiber optic connectors?
- SC and FC
 - RJ-11 and RJ-45
 - VGA and DVI-D
 - DE-9 and DB-25
29. Which of the following is *not* a valid network topology?
- Ring
 - Matrix
 - Mesh
 - Star
30. Which of the following statements accurately describes what the cloud-based service, Infrastructure as a Service (IaaS) provides?
- Provides access to third party cloud-based software.
 - Provides access to resources, such as storage, networking, and virtualization.
 - Provides access to hardware and software tools, commonly used by developers.
 - Provides access to a dedicated, physical location.

31. Which of the following statements accurately describes what the cloud-based service, Platform as a Service (PaaS) provides?
- Provides access to third party cloud-based software.
 - Provides access to resources, such as storage, networking, and virtualization.
 - Provides access to hardware and software tools, commonly used by developers.
 - Provides access to a dedicated, physical location.
32. Which of the following statements accurately describes what the cloud-based service, Infrastructure as a Service (IaaS) provides?
- Provides access to third party cloud-based software.
 - Provides access to resources, such as storage, networking, and virtualization.
 - Provides access to hardware and software tools, commonly used by developers.
 - Provides access to a dedicated, physical location.
33. Which of the following devices / objects would be least likely to be considered part of the Internet of Things (IoT)?
- Smartphones
 - Fitness trackers
 - Medical sensors
 - Printers
34. Which IEEE Standard for Wi-Fi has a theoretical maximum throughput of over 9.6 Gbps?
- 802.11ac
 - 802.11n
 - 802.11g
 - 802.11ax
35. Which generation of broadband cellular network has a theoretical maximum throughput of over 10 Gbps?
- 4G
 - LTE
 - 3G
 - 5G
36. Which of the following accurately describes how account information is stored in a Domain?
- Account information is stored in a centralized database, on a server.
 - Account information is stored in a local database, on each computer.
 - Account information is stored on a shared folder, hosted on a file server.
 - Account information is stored on a network device, such as a switch or router.

37. Which of the following accurately describes how account information is stored in a Workgroup?
- Account information is stored in a centralized database, on a server.
 - Account information is stored in a local database, on each computer.
 - Account information is stored on a shared folder, hosted on a file server.
 - Account information is stored on a network device, such as a switch or router.
38. Which of the following statements is *not true* when considering the primary differences between a proxy server and VPN?
- VPNs encrypt traffic
 - Proxy servers do not encrypt traffic
 - VPNs mask IP addresses
 - Proxy servers encrypt traffic
39. Which of the following statements accurately describes a monitoring filter driver?
- Receives network requests and passes them to other drivers.
 - Receives network requests and inspects (can modify / drop packets).
 - Receives network requests and performs content filtering.
 - Receives network requests and modifies the packet to include a monitoring tag.
40. Users would like to access server resources from home (shared folders, application databases, etc.). Which of the following would be a secure option?
- On the router, forward port 3389 to the server to allow external RDP connectivity.
 - Connect the server directly to the ISP device, bypassing the router, so that it receives an external IP address.
 - Allow users to share a login to third party, remote connectivity software.
 - Use a VPN with encryption enabled.
41. Which of the following types of storage array does *not* provide block-level access to data?
- DAS
 - SAN
 - NAS
 - SaaS
42. A network administrator is troubleshooting an issue in which clients can ping devices on the same network (10.10.1.x), but cannot ping devices on a different network (10.10.2.x). Which of the following is the most likely culprit?
- Router
 - Switch
 - Cables
 - Clients

43. Which of the following security access control methods uses the physical address of the NIC to determine access to a network?
- a. ACL
 - b. VLAN
 - c. IP address blocking
 - d. MAC filtering
44. Which of the following statements is *true* when considering the primary differences between TCP and UDP?
- a. TCP is a connectionless protocol
 - b. TCP is slower than UDP
 - c. UDP is a connection-oriented protocol
 - d. UDP is slower than TCP
45. Which of the following statements accurately describes the practice of multihoming?
- a. Replicating hosts or computer networks between multiple locations.
 - b. Physically moving hosts or computer networks to another location.
 - c. Connecting a host or computer network to itself.
 - d. Connecting a host or computer network to multiple networks.
46. Which of the following is *not* a consideration when implementing Quality of Service (QoS)?
- a. Packet loss
 - b. Throughput
 - c. Jitter
 - d. Cost
47. Which of the following network functions can help with network congestion by limiting the data transmission rate of the sender?
- a. Flow control
 - b. Auto negotiation
 - c. Error control
 - d. Auto sensing
48. Which of the following accurately describes DHCP snooping?
- a. Inspects DHCP requests and forwards them appropriately.
 - b. Aggregates DHCP requests, so that multiple requests can be answered simultaneously.
 - c. Prevents unauthorized DHCP server access and drops unacceptable DHCP traffic.
 - d. Automatically pushes security updates to DHCP enabled clients.

49. When implementing an iSCSI storage network, which of the following is something that would be considered beneficial?
- a. High latency
 - b. Multiple paths
 - c. Low bandwidth
 - d. Single path
50. A network administrator is configuring a storage network to use jumbo frames. Which of the following Windows ping commands, run from the server, could be used to ensure that jumbo frames are functional?
- a. ping <storageIP> -f -l 8972
 - b. ping <storageIP> -f -l 1500
 - c. ping <serverIP> -f -l 8972
 - d. ping <serverIP> -f -l 1500